

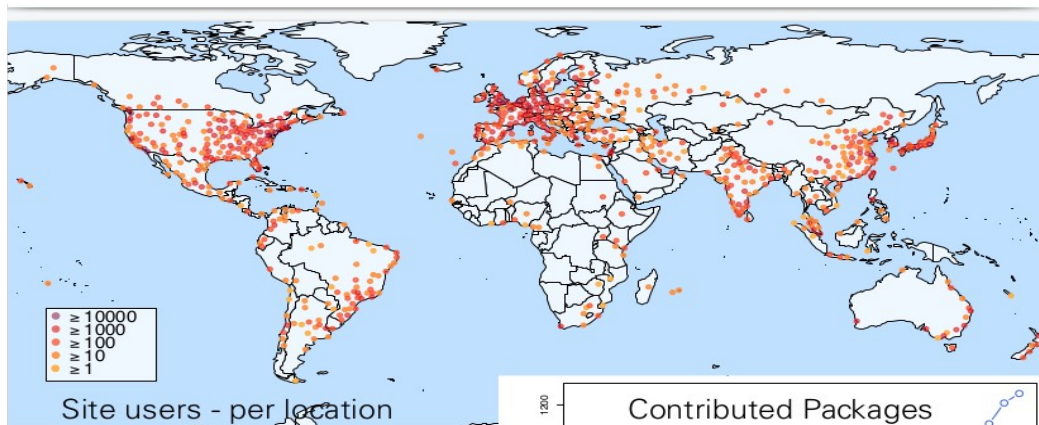
“What marks out modern science is not the conduct of experiments (alchemists conducted plenty of experiments), but the formation of a *critical community capable of assessing discoveries and replicating results.*” The Invention of Science: A New History of the Scientific Revolution, by David Wootton

Alchemy & algorithms: perspectives on the philosophy and history of open science

▼ [Leo Lahti](#), [Filipe da Silva](#), [Markus Petteri Laine](#), [Viivi Lähteenoja](#), [Mikko Tolonen](#)

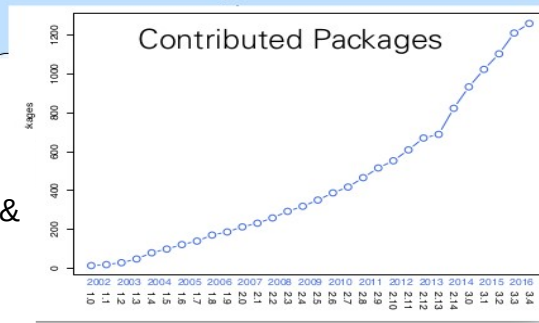


A family of alchemists at work, an engraving by Philip Galle, after a painting by Pieter Bruegel the Elder, published by Hieronymus Cock, c.1558.



World's largest bioinformatics project:

- 2300+ packages
- Tens of thousands of users & publications
- Multi-omics integration



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Perspective | [Published: 02 December 2019](#)

Orchestrating single-cell analysis with Bioconductor

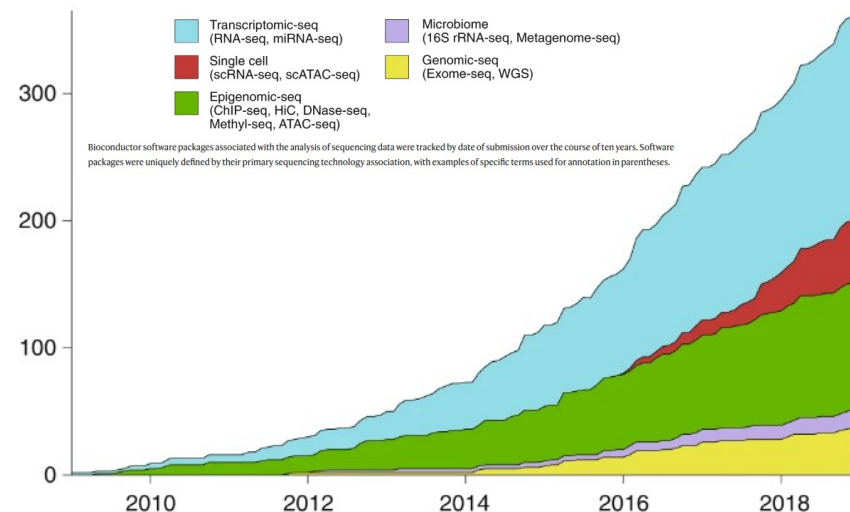
[Robert A. Amezcua](#), [Aaron T. L. Lun](#), [Etienne Becht](#), [Vince J. Carey](#), [Lindsay N. Carpp](#), [Ludwig Geistlinger](#),



Fig. 1: Number of Bioconductor packages for the analysis of high-throughput sequencing data over ten years.

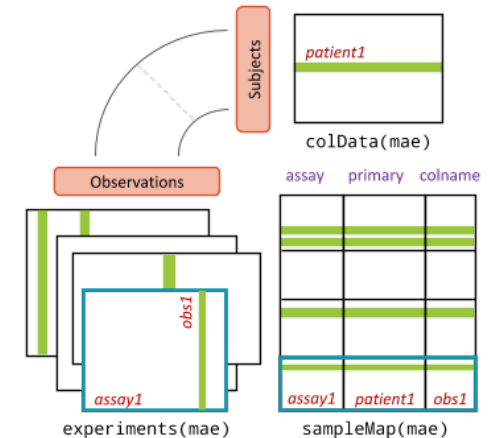
From: [Orchestrating single-cell analysis with Bioconductor](#)

Number of R/Bioconductor packages for the analysis of sequencing data



Data combinations increasingly large & heterogeneous

- Increasing sample sizes, depth & untargeted assays
- Spatio-temporal variation & resolution
- Multi-pipeline (processing, pipelines, transformations)
- Multi-level (hierarchies, phylogenies, nested designs)
- Multi-assay (e.g. 16S + metagenome)
- Multi-omic (taxonomic, functional)
- Multi-kingdom (bacteriome, virome, mykobiome, eukaryome)
- Multi-cohort (meta-analyses)

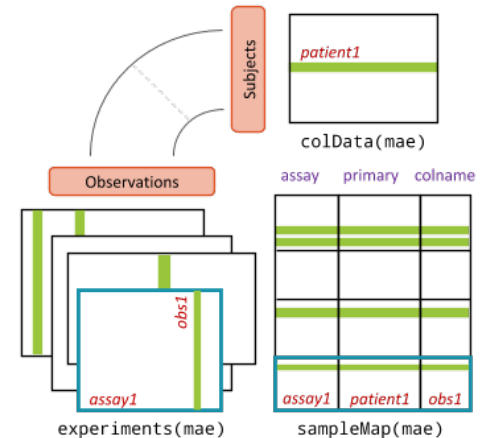


Data combinations increasingly large & heterogeneous

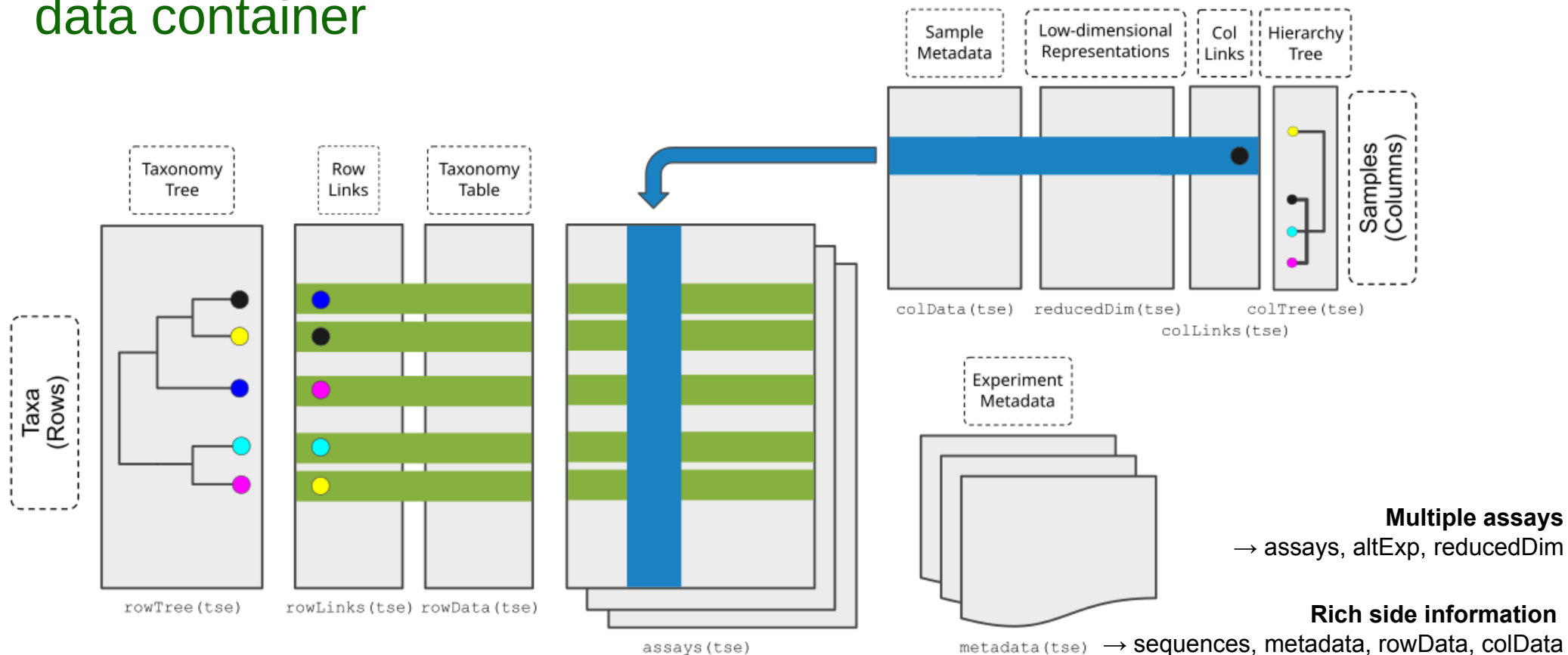
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→ How to organize such data combinations?

→ How to build interoperable methods?



(Tree)SummarizedExperiment data container



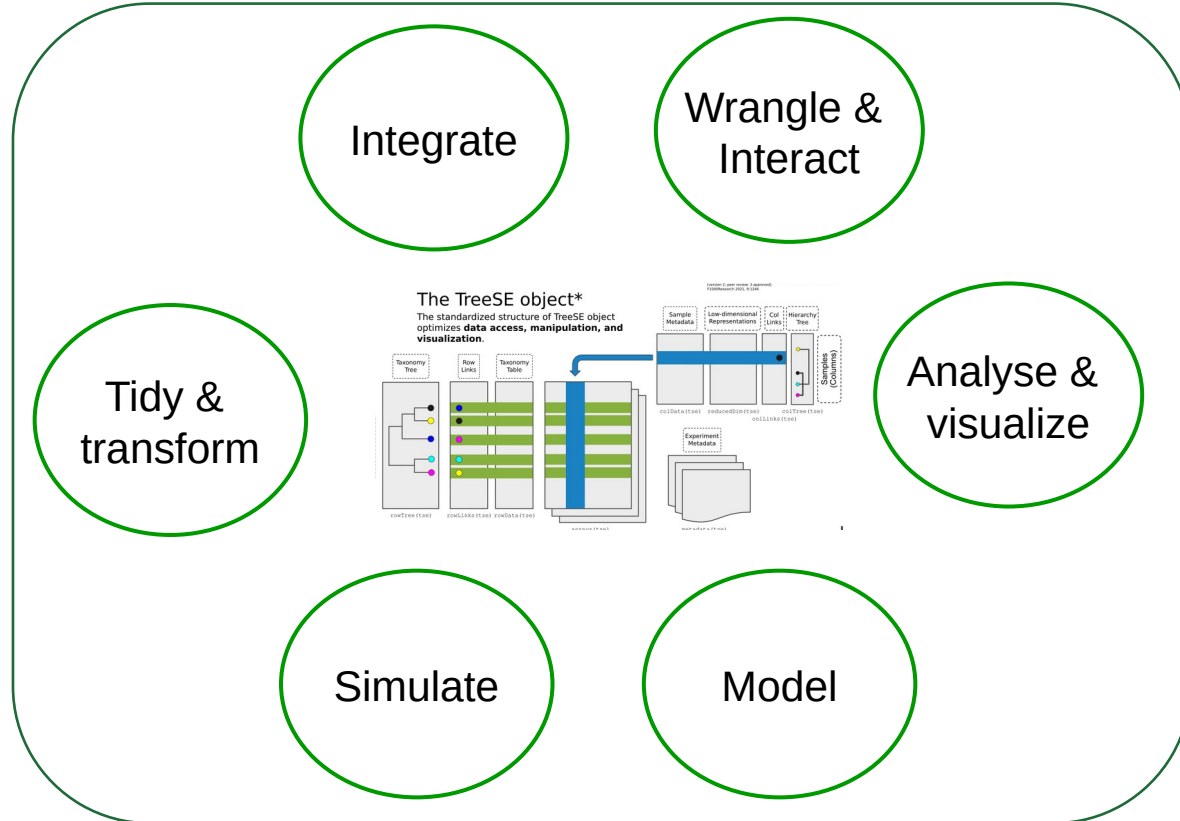
Hierarchies on sample & feature spaces

→ rowTree, colTree

Microbiome data science workflow

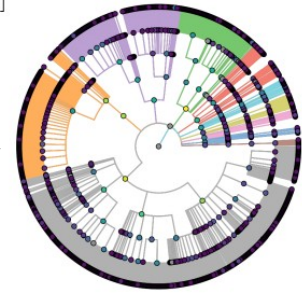
Interoperable methods

Import data



Reproducible reports

[5]



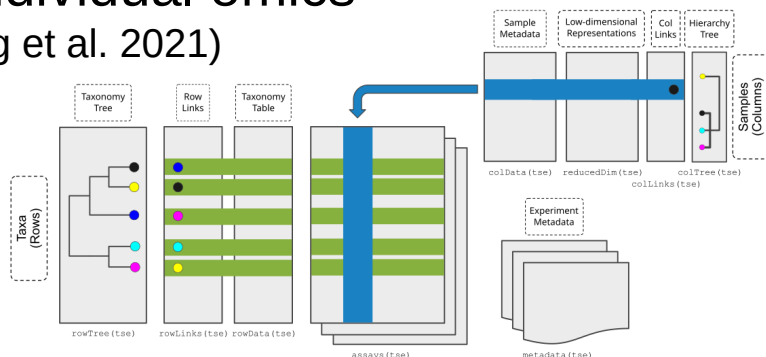
[5] miaViz::plotRowTree(tse)

Novel *data structures* support multi-assay integration & analysis

(Tree)SummarizedExperiment

for individual omics

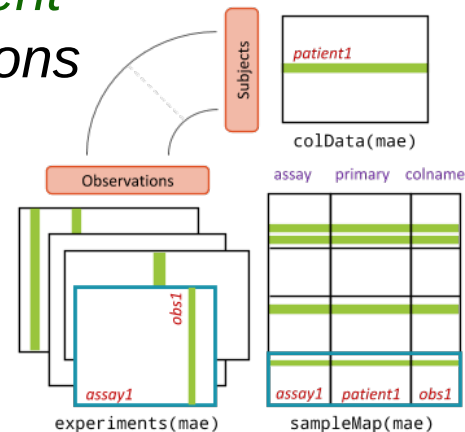
(Huang et al. 2021)



MultiAssayExperiment

for omics combinations

(Ramos et al. 2017)



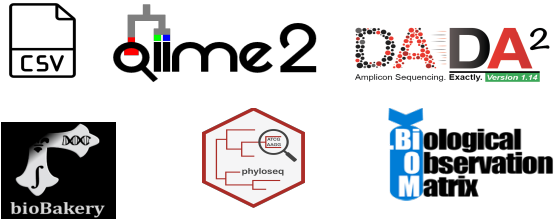
FOCUS ON COMPUTER RESOURCES | NOVEMBER 01 2017

Software for the Integration of Multiomics Experiments in Bioconductor **FREE**

Marcel Ramos; Lucas Schiffer; Angela Re; Rimsha Azhar; Azfar Basunia; Carmen Rodriguez; Tiffany Chan; Phil Chapman; Sean R. Davis; [David Gomez-Cabrero](#); Aedin C. Culhane; Benjamin Haibe-Kains; Kasper D. Hansen; Hanish Kodali; Marie S. Louis; Arvind S. Mer; Markus Riester; Martin Morgan; Vince Carey; Levi Waldron

Easy import for key data formats & resources

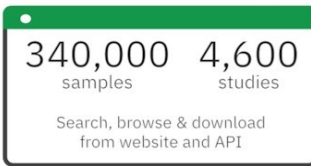
Import common formats:



Retrieve open data:

EBI/MGnify metagenomes, HoloFood, curatedMetagenomicData, ExperimentHub

Database of studies & samples



Simulate microbiome dynamics: miaSim

Methods in Ecology and Evolution

APPLICATION | Open Access | CC BY

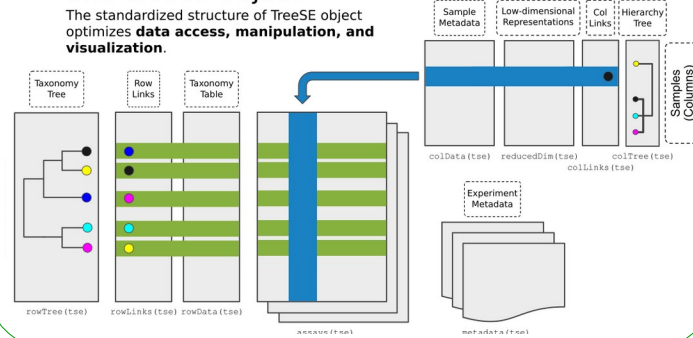
miaSim: an R/Bioconductor package to easily simulate microbial community dynamics

Yu Gao, Yağmur Şimşek, Emma Gheysen, Tuomas Borman, Yi Li, Leo Lahti, Karoline Faust, Daniel Rios Garza

First published: 27 May 2023 | <https://doi.org/10.1111/2041-210X.14129>

The TreeSE object*

The standardized structure of TreeSE object optimizes data access, manipulation, and visualization.

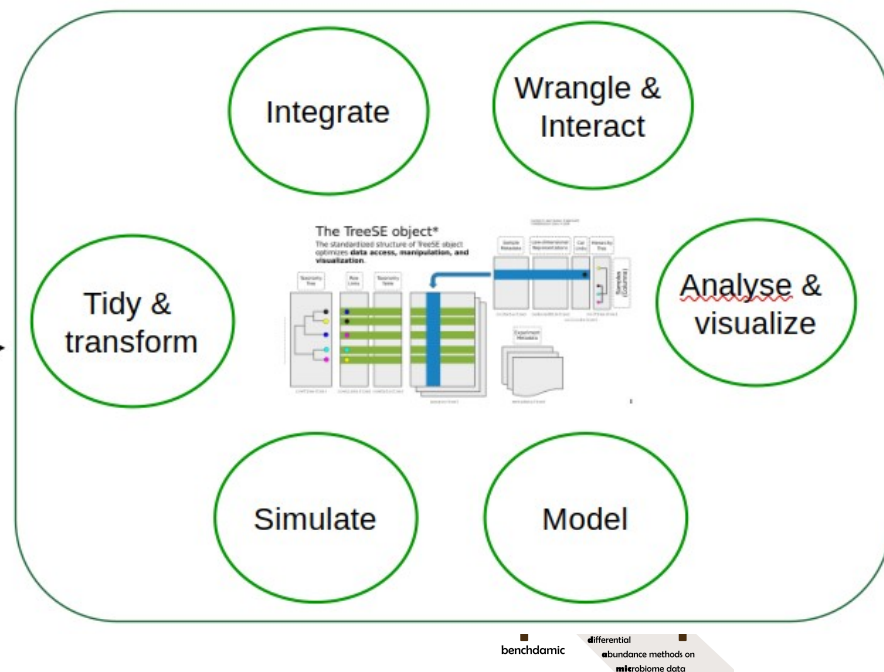


Import into optimized data structures facilitates downstream analyses

Package ecosystem

Custom methods for data import, wrangling, alpha & beta diversity, differential abundance, associations, enrichment, spatial and temporal data, simulation, visualization + interactive applications

- 20+ packages dedicated to microbiome analysis
- 10+ independent, supporting microbiome packages + rest of the R/Bioconductor ecosystem
- Interactive tools (iSEETree)
- 40,000k+ downloads in past 12 months; 500 issues & 1000 PRs closed



The tidyomics ecosystem: enhancing omic data analyses

[William J. Hutchison](#), [Timothy J. Keyes](#), [The tidyomics Consortium](#), [Helena L. Crowell](#), [Jacques Serizay](#), [Charlotte Soneson](#), [Eric S. Davis](#), [Noriaki Sato](#), [Lambda Moses](#), [Boyd Tarlinton](#), [Abdullah A. Nahid](#), [Miha Kosmac](#), [Quentin Clayssen](#), [Victor Yuan](#), [Wancen Mu](#), [Ji-Eun Park](#), [Izabela Mamede](#), [Min Hyung Ryu](#), [Pierre-Paul Axisa](#), [Paulina Paiz](#), [Chi-Lam Poon](#), [Ming Tang](#), [Raphael Gottardo](#), [Martin Morgan](#), ... [Stefano Mangiola](#) ✉ [+ Show authors](#)

[Nature Methods](#) 21, 1166–1170 (2024) | [Cite this article](#)



Orchestrating Microbiome Analysis



Online Gitbook (OMA) - beta version

Introduction to the new data science framework

- 1) Downstream analysis
- 2) Statistical programming
- 3) Alternative to *phyloseq*

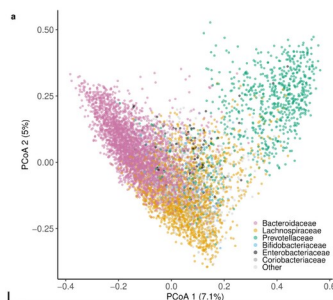
Online support & training workshops:
Pune (IND) | Turku & Oulu (FIN) | Utrecht, Wageningen, Radboud (NLD) | Norwich (UK) | Bergen (NO), ECCB (FIN), EuroBioC (UK)

Population variation and prognostic potential of gut antibiotic resistance

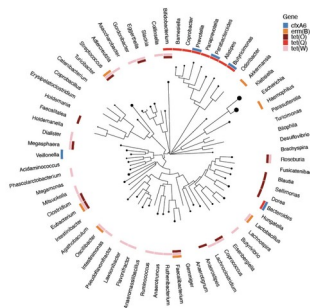
👤 K. Pärnänen, 👤 M. Ruuskanen, 👤 G. Sommeria-Klein, 👤 V. Laitinen, 👤 P. Kantanen, 👤 G. Méric, 👤 C. Gazolla Volpiano, 👤 M. Inouye, 👤 R. Knight, 👤 V. Salomaa, 👤 A. S. Havulinna, 👤 T. Niiranen, L. Lahti

doi: <https://doi.org/10.1101/2024.08.08.24311663>

`runMDS()`

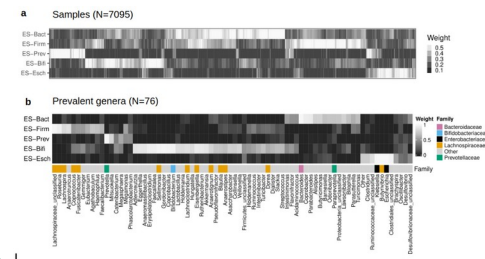
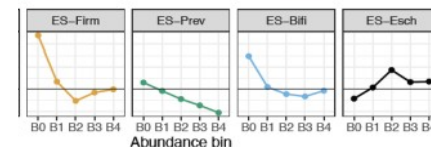


Bacterial phylogeny vs. resistance genes:
`plotFeatureLoadings()`



Enterosignatures vs. ARG load:

`agglomerateByRank()` → `getNMF()`



Thanks!

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Department of Computing, University of Turku, Finland

40+ contributors from 10+ countries

Giulio Benedetti, Geraldson Muluh, Akewak Jeba, Basil Courbayre, Renuka Potbhare, Himmi Lindgren, Ely Seraidarian, Yağmur Şimşek, Théotime Pralas, Noah de Gunst, Axel Dagnaud, Sam Hillman, Matti Ruuskanen, Katariina Pärnänen, Pande Erawijantari, Aura Raulo, Mahkameh Salehi, Vilhelm Suksi, Kevin Gatel, Teo Dallier, Elliot Gaudron-Barry, Inès Benseddik, Shadman Ishraq, Jiya Chaudhary, Elina Chiesa, Shigdel Rajesh, Artur Sannikov, Henrik Eckermann, Lu Yang, Stefanie Peschel, Aaron Lun, Levi Waldron, Matteo Calgaro, Vivian Ikeh, Moritz Beber, Jacques Serizay, Karoline Faust, Daniel Garza, Yang Cao, Eineje Ameh, Sudarshan Shetty, Ruizhu Huang, Danielle Callan, Yu Gao, Ben Allen, Jesse Pasanen, Felix Ernst, Domenick Braccia, Héctor Corrada Bravo, Himel Mallick

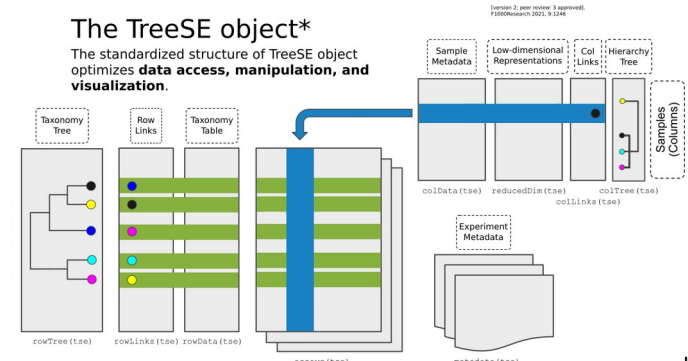
Independent package developers: scater, scuttle, MicrobiotaProcess, pHILR, ANCOMBC, ALDEx2, benchdamic, (Tree)SummarizedExperiment, SingleCellExperiment, MultiAssayExperiment, vegan, curatedMetagenomicData, ExperimentSubset, CBEA, microSTASIS

microbiome.github.io



The TreeSE object*

The standardized structure of TreeSE object optimizes **data access, manipulation, and visualization**.



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